THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 9: **Methods of Separation in Everyday Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student wants to dry sea salt. She should: 1

(a) add water **(b)** filter
(c) stir **(d)** spread and heat it in sun

2. A solid dissolves more in hot water than cold. A saturated hot solution when cooled will: 1

(a) deposit crystals **(b)** evaporate **(c)** become clear **(d)** become hotter

3. Which of these mixtures needs both a magnet and water to separate all its parts? 1

(a) sand and gravel **(b)** iron filings, salt and sand
(c) rice and husk **(d)** salt and water

4. In a lab, separation of a mixture of salt and water must keep the salt. Which method is correct? 1

(a) evaporation **(b)** sieving **(c)** winnowing **(d)** decantation

5. A student tries to separate a salt solution by filtration and gets salt in the filtrate. This shows that: 1

(a) salt passes through filter paper **(b)** salt is insoluble
(c) it is not a solution **(d)** filter paper is damaged

6. Which of these does not need heating at any stage? 1

(a) separating iron by magnet **(b)** evaporation
(c) crystallisation **(d)** distillation

7. Which does not involve a change of state? 1

(a) filtration **(b)** evaporation **(c)** condensation **(d)** distillation

8. Which of the following indicates a successful filtration? 1

(a) no filtrate **(b)** coloured residue with all liquid
(c) clear filtrate **(d)** cloudy filtrate

9. **Assertion (A):** Churning of curd gives butter. 1
Reason (R): Churning separates the lighter fat from the other parts of curd or milk.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Iron nails can be separated from a mixture with sand using a magnet. 1
Reason (R): Sand is attracted by a magnet.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the filtrate remain cloudy if the filter paper tears? 2

12. Explain why hot saturated salt solution gives crystals on cooling. 2

13. Why is filtration useful for muddy water but not for salt water? 2

14. Why can't dissolved gases in water be separated by filtration? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why people separate cream from milk and what the separated liquid is. 3

16. Why is it necessary to dry the residue after filtration? 3

17. Name the principle used in a washing machine's drain filter. 3

OR

A student mixes sugar, chalk and water. How will he get both sugar and chalk? 3

18. Describe the separation of mustard oil from mustard seeds in a simple way. 3

OR

Why do food industries use sieves and magnets? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** At a salt farm near the coast of Gujarat, sea water is let into shallow pans. Under the hot sun the water slowly disappears, leaving white salt crystals. In a school lab, Hari does the same with salty water in a dish, then tries to collect the vapour on a cool plate held above. 5
(i) Name the process that leaves salt behind.
(ii) Name the change of vapour into drops on the cool plate.
(iii) Which solid is left in the pans?
(iv) If Hari wants the water as well as the salt, how can he get both?

20. Explain with an example why a method of separation that works for one mixture may not work for another. 5

OR

Explain how a scientist would test the purity of water from a pond, a tap and distilled water using evaporation. 5

END OF QUESTION PAPER — Total: 40 Marks